IT PROJECT MANAGEMENT

Comprehensive Course Reviewer (Weeks 2 – 7)

WEEK 2: INTRODUCTION TO PROJECT MANAGEMENT

1. Overview & Core Definitions

Project management provides a structured methodology for planning, executing, monitoring, and completing temporary endeavors aimed at delivering unique outcomes.

> **Definition of Project Management:** *"The application of knowledge, skills, tools, and techniques to project activities to meet project requirements."* — **PMBOK Guide (7th Edition)**
>
> *"Project management is a structured approach to planning, executing, monitoring, and closing temporary efforts to deliver a unique product or service."* — **Portny (2020)**

2. Importance of Project Management in IT

IT projects operate in high-risk, rapidly evolving environments characterized by complex technical architectures, tight execution timelines, and strict resource constraints. Proper project management is critical due to:

- **High Failure & Overrun Rates:** Mitigates frequent budget overruns, missed deadlines, and technical delivery failures.
- **Adaptive Planning Needs:** Enables integration of flexible frameworks such as *Agile* and *DevOps* to match technical dynamism.
- **Cross-Functional Coordination:** Facilitates seamless collaboration among technical teams, business stakeholders, and external vendors.
- **Real-World Examples:** Developing a secure mobile banking app, migrating enterprise database systems to the cloud, and implementing Enterprise Resource Planning (ERP) solutions.

3. Key Roles and Responsibilities in IT Projects

ROLE	KEY RESPONSIBILITIES
Project Manager (PM)	Oversees end-to-end project execution, including scope, budget, schedule, risk management, stakeholder alignment, and communication.
System Analyst	Gathers, formalizes, and translates business requirements into precise technical specifications; performs feasibility analyses.
Developers / Programmers	Construct, code, debug, and unit-test software components based on architectural and design requirements.
UI/UX Designers	Design intuitive, accessible, and user-centric interfaces and user experience workflows.
QA Testers	Perform rigorous functional, integration, non-functional, security, and user acceptance testing to guarantee quality.
Network / Infra Engineers	Manage system deployment, hardware provisioning, cloud infrastructure setup, network protocols, and environment configuration.
Stakeholders	Internal or external individuals or groups (sponsors, clients, end-users) with a direct interest in or impact on the project outcome.

4. Standards & Governance: PMI and the PMBOK Guide

- **Project Management Institute (PMI):** A globally recognized professional non-profit organization dedicated to establishing standards, credentials, and best practices in project management (e.g., PMP, CAPM).
- **PMBOK Guide (7th Edition) Performance Domains:**
  1. **Stakeholders:** Productive working relationships with all stakeholders.
  2. **Team:** High-performing team dynamics and leadership development.
  3. **Development Approach & Life Cycle:** Aligning cadence and delivery models.
  4. **Planning:** Initial and ongoing alignment of project scope and tasks.
  5. **Project Work:** Managing physical resources, procurement, and process efficiency.
  6. **Delivery:** Fulfilling requirements and achieving expected business outcomes.
  7. **Measurement:** Evaluating performance against baselines using metrics.
  8. **Uncertainty:** Identifying, evaluating, and addressing risks and ambiguity.

5. Common IT Project Challenges & Best Practices

Major Challenges:

- **Scope Creep:** Uncontrolled expansion of project scope without adjustments to time, cost, or resources.
- **Poor Communication:** Disconnect and misalignment between technical delivery teams and non-technical business stakeholders.
- **Unrealistic Deadlines:** Arbitrary delivery target dates set without technical estimation or complexity evaluation.
- **Inadequate Risk Management:** Failure to identify or mitigate technical, compliance, and security vulnerabilities early.
- **Technology Volatility:** Sudden shifts in frameworks, platform dependencies, security standards, or software tools.
- *21st Century PM Research (Tabassi et al., 2019):* Highlights 18 main challenges across 125 parameters, stressing that traditional PM frameworks often lack soft-skills coverage, requiring manager flexibility and organizational learning.

Proven Best Practices for IT PM Success:

- Establish clear, validated, and unambiguous requirements during initiation.
- Select appropriate methodologies (e.g., *Agile* or *Hybrid*) adapted to project parameters.
- Maintain continuous, transparent stakeholder engagement and reporting.
- Conduct structured, recurring risk assessments throughout the project lifecycle.
- Invest heavily in centralized communication platforms and complete project documentation.

WEEK 3: PROJECT LIFECYCLE AND METHODOLOGIES

1. Overview of the IT Project Lifecycle

The *IT Project Lifecycle* represents the sequential and overlapping phases a project traverses from inception to final completion. Tailored specifically to project size and complexity (Nizhebetskyi, 2022).

LIFECYCLE PHASE	CORE PURPOSE & ACTIVITIES	EXAMPLE (E-LEARNING PORTAL)
1. Initiation	Define project objectives, evaluate feasibility, identify stakeholders, and secure formal approval.	Identify organizational need to digitize course delivery and evaluate platform options.
2. Planning	Detail technical scope, establish schedules, allocate budgets, set quality baselines, and plan risk responses.	Allocate capital budget, design infrastructure architecture, and select LMS tools.

LIFECYCLE PHASE	CORE PURPOSE & ACTIVITIES	EXAMPLE (E-LEARNING PORTAL)
3. Execution	Mobilize resources, build software components, design interfaces, write backend code, and configure systems.	Develop frontend portal interfaces and code backend database integration.
4. Monitoring & Control	Track Key Performance Indicators (KPIs), verify quality standards, manage schedule variance, and process changes.	Track usability testing feedback, monitor code defect rates, and control feature requests.
5. Closure	Conduct final user acceptance deployment, complete documentation, execute handover, and perform post-project review.	Final production deployment, operational staff training, and system operational handover.

2. Comparative Methodology Analysis: Waterfall, Agile, and Hybrid

FEATURE / DIMENSION	TRADITIONAL (WATERFALL)	AGILE APPROACH	HYBRID METHODOLOGY
Core Nature	Linear & Sequential; distinct gated phases.	Iterative & Incremental; continuous refinement.	Blended; structured governance + iterative execution.
Flexibility	**Low:** Changes midstream are costly and difficult.	**High:** Embraces changing scope dynamically.	**Medium:** Balanced adaptiveness within set boundaries.
Feedback Loop	**Late:** Stakeholders evaluate product at closure.	**Continuous:** Regular demos and sprint reviews.	**Balanced:** Key milestone checks and sprint reviews.
Project Type Fit	Stable, highly predictable scope (e.g., hardware/db migration).	Evolving requirements, rapid innovations (e.g., mobile apps).	Regulated environments requiring compliance + rapid development.
Documentation	**Extensive:** Heavily documented requirements prior to build.	**Minimal:** Prioritizes working software over documentation.	**Moderate:** Standard compliance docs with lean execution logs.
Delivery Speed	**Slower:** Value delivered only at end of timeline.	**Fast:** Frequent releases of functional increments.	**Moderate:** Phased releases linked to major milestones.

3. The Scrum Framework

Scrum is a lightweight Agile framework for delivering products in 2- to 4-week timeboxed iterations called *Sprints* (Schmidt, 2021).

Core Scrum Roles:

- **Product Owner:** Represents business value, defines User Stories, maintains and prioritizes the *Product Backlog*, and manages stakeholder expectations.
- **Scrum Master:** Process facilitator and servant-leader; removes organizational blockers, enforces Scrum rules, and shields the team.
- **Development Team:** Self-organizing, cross-functional group responsible for delivering a potentially releasable product increment each sprint.

Key Scrum Events (Ceremonies):

- **Sprint Planning:** Collaborative alignment event to define the *Sprint Goal* and select backlog items.

- **Daily Scrum (Stand-up):** Brief daily 15-minute synchronization meeting to discuss progress and impediments.
- **Sprint Review:** Demonstration of the completed functional software increment to key stakeholders.
- **Sprint Retrospective:** Post-sprint internal team evaluation focused on continuous process improvement.

WEEK 4: PROJECT INTEGRATION AND INITIATION

1. Project Integration Management

Portny (2020) defines *Project Integration Management* as coordinating all project elements—scope, time, cost, quality, resources, and communications—to ensure unified execution.

> **Strategic Importance: Nizhebetskyi (2022)** emphasizes integration as the *"strategic glue"* keeping projects aligned with their business purpose and preventing siloed execution across functional units.

Key Processes Under Integration Management (PMBOK Aligned):

1. **Develop Project Charter:** Formally authorizing a project and giving authority to the PM.
2. **Develop Project Management Plan:** Consolidating subsidiary plans into a master strategy.
3. **Direct and Manage Project Work:** Leading team execution of planned activities.
4. **Monitor and Control Project Work:** Tracking, reviewing, and regulating progress against plan.
5. **Perform Integrated Change Control:** Reviewing, approving, and managing changes to baselines.
6. **Close Project or Phase:** Formally finalizing all activities, handovers, and contracts.

2. Purpose and Core Steps of Project Initiation

Initiation formally authorizes a project and sets high-level goals (Dumitrascu, 2021). **Schmidt (2021)** views it as the *"decision-making gateway."*

Essential Initiation Activities:

- **Business Case Development:** Justifying why the project is important and what value it delivers.
- **Feasibility Assessment:** Evaluating technical, financial, and logistical viability.
- **Stakeholder Identification:** Determining who is affected, who approves, and who funds the project.
- **Project Charter Creation:** Authoring the foundational document that launches the project lifecycle.

3. Essential Components of a Project Charter

CHARTER COMPONENT	DESCRIPTION & EXEMPLAR DETAILS
1. Project Title	Concise, formal name (e.g., *Responsive Student Portal System*).
2. Purpose / Justification	High-level rationale (e.g., *Streamline enrollment and reduce manual processing errors*).
3. Objectives & Success Criteria	SMART goals (e.g., *Launch by Q4 2026; achieve 95% user satisfaction post-launch*).
4. High-Level Requirements	Core deliverables (e.g., *Mobile accessibility, user dashboard, automated reporting*).
5. Summary Milestone Schedule	Target dates for major phases (e.g., *Design: Aug; Beta: Oct; Deployment: Nov*).
6. Budget Overview	Authorized funding allocation (e.g., *₱500,000 for development, testing, and training*).
7. Key Stakeholders & Roles	Designated leaders (e.g., *Project Manager: Ms. Cruz; Sponsor: IT Director*).

CHARTER COMPONENT	DESCRIPTION & EXEMPLAR DETAILS
8. Assumptions & Constraints	Operational parameters (e.g., *Assumption: Users have internet access; Constraint: Limited training time*).
9. Authorization Section	Official sign-off from project sponsor empowering the PM.

WEEK 5: PROJECT SCOPE MANAGEMENT

1. Role and Importance of Project Scope Management

Project Scope Management refers to the systematic process of defining, documenting, and controlling what is and what is not included in a project. It ensures that an IT project includes all required work—and *only* required work.

Core Roles of Scope Management:

- **Define Boundaries:** Clarifies deliverables and explicitly states exclusions (e.g., LMS Version 1 includes quizzes but excludes AI chatbots).
- **Guide Planning:** Provides the foundation for schedules, budgets, and resource allocation.
- **Align Stakeholders:** Acts as a contract between developers, management, and clients to manage expectations.
- **Control Changes:** Establishes formal change request mechanisms.

Importance in IT Project Success:

- **Prevents Scope Creep:** Blocks unapproved, incremental feature requests ("just one more feature") from inflating budgets and delaying timelines.
- **Improves Resource Management:** Ensures technical talent is spent only on agreed scope items.
- **Ensures Quality:** Sets explicit acceptance criteria for deliverables.

2. Six Core Processes of Scope Management (PMBOK Aligned)

1. **Plan Scope Management:** Creating a written Scope Management Plan detailing how scope will be defined, validated, and controlled (Dumitrascu, 2021).
2. **Collect Requirements:** Gathering stakeholder needs via interviews, surveys, and workshops.
3. **Define Scope:** Developing a detailed Project Scope Statement outlining objectives, deliverables, boundaries, assumptions, and constraints.
4. **Create Work Breakdown Structure (WBS):** Decomposing total project work into smaller, manageable components (work packages).
5. **Validate Scope:** Formalizing acceptance of completed project deliverables with stakeholders.
6. **Control Scope:** Monitoring project status and managing changes to the scope baseline to prevent scope creep.

3. Components of a Scope Baseline & Scope Management Plan

The **Scope Baseline** consists of three key elements: (1) *Project Scope Statement*, (2) *WBS*, and (3) *WBS Dictionary*.

Example Scope Baseline Elements (Learning Management System - LMS):

- **Project Overview & Objectives:** Develop a secure web/mobile LMS in 8 months under PHP 5M budget.
- **Scope Description (Inclusions):** User management, course creation, automated quiz grading, analytics dashboard, payment gateway API, mobile responsiveness.
- **Project Boundaries (Exclusions):** Advanced AI tutoring, social media integration, offline mode without internet.
- **Constraints & Assumptions:** PHP 5M budget, compliance with Data Privacy Act (RA 10173); Assumption of steady internet availability for end users.

- **Acceptance Criteria:** Support 10,000 concurrent users, 99.5% uptime, 100% UAT pass rate.

WEEK 6: PROJECT TIME MANAGEMENT AND SCHEDULING

1. Concepts & Definitions

- **Project Management Timeline:** A visual schedule breaking the project into tasks and milestones with assigned deadlines to anticipate bottlenecks.
- **Schedule vs. Timeline:** A *Schedule* is a granular to-do list (tasks, dates, assignees), whereas a *Timeline* provides a high-level overview of key milestones and overall sequence.
- **Core Timeline Elements:** Specific tasks, start/end dates, task duration, dependencies, and assigned resources.

2. Visualization Tools: Gantt Chart vs. PERT Chart

FEATURE	GANTT CHART	PERT CHART
Visual Format	Horizontal bar chart (tasks on Y-axis, timeline on X-axis).	Network diagram (nodes = tasks/ milestones, arrows = dependencies).
Primary Focus	Task timelines, resource allocation, and progress tracking.	Task dependencies, logical sequencing, and critical path.
Time Estimation	Single-point estimates against calendar dates.	Three-point estimates (Optimistic, Pessimistic, Most Likely).
Best Suited For	Well-defined projects in progress; stakeholder reporting.	Complex, large-scale planning phase projects (R&D, systems).
Progress Tracking	Excellent (shaded bars show % completed).	Limited (focuses on milestone achievement, not % complete).

3. Developing a Schedule & Critical Path Method (CPM)

Developing a schedule requires breaking down work (WBS), identifying task dependencies, estimating durations, and establishing deadlines and milestones.

CPM Network Variables:

- **Earliest Start (ES):** Earliest time a task can begin.
- **Earliest Finish (EF):** $EF = ES + t$ (where t is activity duration).
- **Latest Finish (LF):** Latest time a task can finish without delaying project completion.
- **Latest Start (LS):** $LS = LF - t$.
- **Slack / Float:** $Slack = LF - EF$ (or $LS - ES$). Tasks on the Critical Path have **Zero Slack**.

CPM Two-Pass Algorithm:

- **Forward Pass (Left-to-Right):** Calculates ES and EF for all activities. $ES = ext{max}(EF ext{ of predecessors})$. The largest EF indicates overall project duration.
- **Backward Pass (Right-to-Left):** Calculates LF and LS starting from project duration. $LF = ext{min}(LS ext{ of successors})$.
- **Critical Path Identification:** The continuous sequence of dependent tasks where $Slack = 0$. Any delay in critical path tasks delays the entire project. Tasks with positive slack can be delayed without affecting completion date.

WEEK 7: PROJECT COST MANAGEMENT

1. Fundamentals of Cost Management

Project Cost Management involves planning, estimating, budgeting, and controlling costs to complete a project within the approved budget.

Five Core Cost Components:

- **Direct Costs:** Expenses directly tied to project work (e.g., developer salaries, dedicated software fees).
- **Indirect Costs:** Shared operational expenses (e.g., utility bills, facility rentals).
- **Fixed Costs:** One-off costs unlinked to project duration (e.g., hardware server purchase).
- **Variable Costs:** Costs fluctuating with time or volume (e.g., contractor hourly fees, cloud consumption).
- **Sunk Costs:** Unrecoverable past expenditures that should not influence future decisions.

2. Cost Management Plan & Functions

Four primary functions of cost management: (1) *Resource Planning* (using Resource Breakdown Structure - RBS), (2) *Cost Estimating*, (3) *Cost Budgeting* (establishing baseline + contingency), and (4) *Cost Control*.

Estimation & Pricing Models:

- **Top-Down Estimating:** High-level estimation based on historical data; useful early when scope is high-level.
- **Bottom-Up Estimating:** Estimating detailed individual tasks (WBS/CBS) and summing them up; highly accurate.
- **Pricing Calculations:** Hourly (rate $ imes$ hours), Flat Rate (fixed price upfront), Cost-Plus (actual cost + % margin), Value-Based (charged on realized client benefit).

3. Earned Value Management (EVM) & Formulas

EVM is a standard performance measurement technique integrating scope, schedule, and cost metrics.

- **Earned Value (EV) = Budgeted Cost of Work Completed = Total Task Budget × % Completed**
- **Actual Cost (AC) = Actual money spent for work performed up to date**
- **Planned Value (PV) = Budgeted Cost of Work Scheduled up to date**
- **Cost Variance (CV) = EV – AC (Positive = Under Budget | Negative = Over Budget)**
- **Schedule Variance (SV) = EV – PV (Positive = Ahead of Schedule | Negative = Behind Schedule)**
- **Cost Performance Index (CPI) = EV / AC (CPI ≥ 1: Efficient | CPI < 1: Inefficient/Over Budget)**
- **Schedule Performance Index (SPI) = EV / PV (SPI ≥ 1: Ahead/On Schedule | SPI < 1: Behind Schedule)**
- **Estimate at Completion (EAC) = BAC / CPI (Projected total cost upon project completion)**

Worked EVM Numerical Example (Web-Based LMS Project):

Baseline Setup: Total Budget at Completion (BAC) = ₱45,000. Month 1 Planned Value (PV) = ₱10,000.
Month 1 Execution Status: Actual Cost (AC) spent = ₱12,000. Completed Work = 80% of Month 1 plan.

EVM METRIC	CALCULATION & VALUE	INTERPRETATION
Earned Value (EV)	₱10,000 $ imes$ 0.80 = **₱8,000**	Monetary value of work actually completed.
Cost Variance (CV)	8,000 – 12,000 = **-₱4,000**	**Over Budget:** Spent ₱4,000 more than earned value.
Schedule Variance (SV)	8,000 – 10,000 = **-₱2,000**	**Behind Schedule:** Accomplished ₱2,000 less value than planned.
Cost Performance Index (CPI)	8,000 / 12,000 = **0.67**	**Cost Inefficiency:** Getting only ₱0.67 value per ₱1.00 spent.

EVM METRIC	CALCULATION & VALUE	INTERPRETATION
Schedule Performance Index (SPI)	8,000 / 10,000 = **0.80**	**Schedule Delay:** Progressing at 80% of planned pace.
Estimate at Completion (EAC)	45,000 / 0.67 = **₱67,164**	**Forecasted Total Cost:** Project will overrun budget by ₱22,164 if trend continues.

REFERENCES

Dumitrascu, S. (2021). *IT Project Management Essentials: A Practical Guide*. Independently published.

Layton, M. C., & Ostermiller, S. J. (2020). *Agile project management for dummies*. John Wiley & Sons.

Nizhebetskyi, D. (2022). *Practical project management: Proven framework that great project managers use in the real world*. Dmytro Nizhebetskyi.

Portny, S. E. (2020). *Project management all-in-one for dummies*. John Wiley & Sons.

Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)* (7th ed.). Project Management Institute.

Schmidt, T. (2021). *Strategic project management made simple: Solution tools for leaders and teams*. John Wiley & Sons.

Tabassi, A. A., Bryde, D., Kamal, E. M., & Dowson, J. (2019). *Challenges for project management in the 21st century*. European Proceedings of Social and Behavioural Sciences.